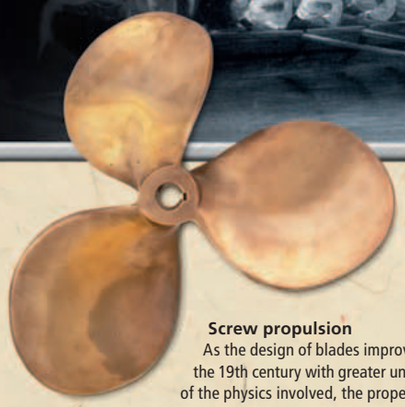




Screw propulsion
As the design of blades improved through the 19th century with greater understanding of the physics involved, the propeller totally superseded the paddle wheel.

combined naval strength of France and Russia led Britain to formally adopt the "two-power standard." The Naval Defence Act of 1889 decreed that the Royal Navy should always be strong enough to defeat any other two navies, and pledged vast sums of public money to the construction of new battleships and cruisers. But Germany was to provide the real challenge to British naval superiority from the 1890s, when it placed its expanding industrial power behind a massive shipbuilding programme under Admiral Alfred von Tirpitz.

IMPERIAL ASSERTION

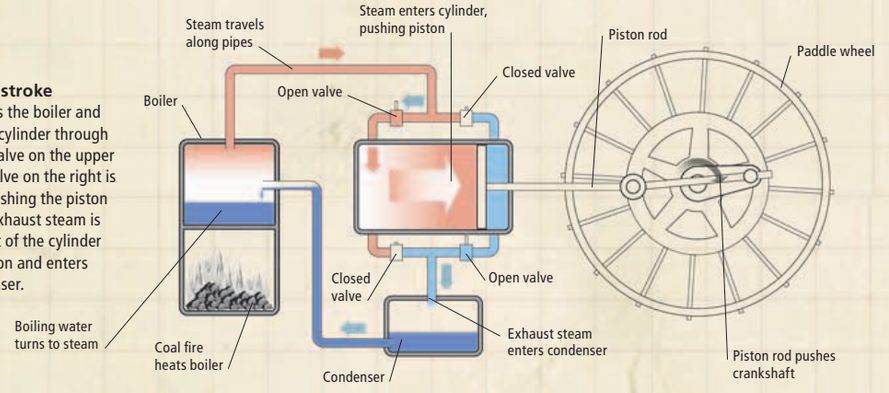
Sea travel became far more secure in the second half of the 19th century after the great powers formally renounced privateering and most piracy was quelled. As the arm of the world's dominant shipbuilding and commercial power, the Royal Navy took the lead in charting the seas—a major project that led to safer navigation for all—and the suppression of the slave trade. But European dominance of the oceans was most often experienced by the rest of the world as straightforward bullying. This was seen at its worst in the mistreatment of China by Britain and France from the 1840s to the 1880s. As long as European powers did not fight one another, however, there was little chance for the expensive steam fleets to show their worth in battle.

WEAPONS AND TECHNOLOGY

EARLY NAVAL STEAM ENGINES

The standard propulsion system for early steamships was a paddle wheel driven by a double-acting steam engine (see below). Steam entered the cylinder alternately on the left and right, pushing the piston forward and back. The first ironclad battleships—the French *Gloire* of 1859 and British *Warrior* of 1861—combined double-acting engines with screw propellers. In the 1880s the introduction of triple-expansion engines, with three cylinders instead of one, led to a major improvement in efficiency. Whereas *Warrior*'s engine needed 5 lb (2.2 kg) of coal to generate one horsepower for an hour, the triple-expansion engine of the British pre-dreadnought *Canopus* in 1899 needed only 1.7 lb (0.75 kg). In the early 20th century Charles Parsons's steam turbine would take steam propulsion to its peak of efficiency.

Forward stroke
Steam exits the boiler and enters the cylinder through the open valve on the upper left (the valve on the right is closed), pushing the piston forward. Exhaust steam is pushed out of the cylinder by the piston and enters the condenser.



Backward stroke
The valve on upper left closes and the valve on the right opens, so the steam now enters the cylinder at the far end and pushes the piston back. The exhaust steam now enters the condenser through the open valve on the left.

